Grade 6 Pre-Assessment

Do your best. It's okay to skip a problem you don't know yet.

Mastery: 28 of 35 (80%)

Part 1 Ratios

1) $6 : 1 = 24 : 4$

2) $9 : 2 = 27 : 6$

Part 2 Rates

3) 150 miles in 5 hours = 30 miles per hour

4) 72 pages in 6 days = 12 pages per day

Part 3 Percents

5) 20% of 85 = 17

6) 25% of 44 = 11

Part 4 Dividing fractions

7) $\frac{5}{8} \div \frac{1}{4} = \frac{5}{2}$

8) $\frac{2}{3} \div \frac{1}{9} = 6$

Part 5 Long division

9) $3{,}404 \div 23 = 148$

10) $6{,}370 \div 26 = 245$

Part 6 Decimal operations

11) $16.2 \div 0.9 = 18$

12) $9.75 \div 1.5 = 6.5$

Grade 6 Pre-Assessment (continued)

Part 7 GCF and LCM

13) GCF of 50 and 80 = 10

14) LCM of 4 and 7 = 28

Part 8 Negative numbers Write < or >.

15) 5 > −5

16) −2 > −12

Part 9 Expressions

17) when n = 3: 7n + 8 = 29

18) when n = 10: 4n + 3 = 43

Part 10 One-step equations

19) x + 26 = 43: x = 17

20) 9x = 99: x = 11

Part 11 Area of shapes

21) triangle, base 13 in, height 8 in: area = 52 sq in

22) triangle, base 9 m, height 6 m: area = 27 sq m

Part 12 Volume with fractions

23) $1\frac{1}{2} \times 3 \times 3 = 13\frac{1}{2}$

24) $2\frac{1}{2} \times 2 \times 6 = 30$

Part 13 Surface area

25) box 5 × 3 × 4 cm: surface area = 94 sq cm

Grade 6 Pre-Assessment (continued)

Part 13 Surface area (continued)

26) box 9 × 4 × 2 ft: surface area = **124** sq ft

Part 14 Mean and median

27) median of 7, 15, 9, 13 = **11**

28) mean of 5, 8, 6, 9 = **7**

Part 15 Data

29) 12, 13, 13, 14, 48: mean = **20** and median = **13**

Part 16 Word problems

30) A car goes 336 miles in 6 hours. How many miles does it go in 1 hour?

336 ÷ 6 = 56

Answer **56** miles

31) Juice is mixed 2 cups of mango for every 5 cups of water. How much water goes with 6 cups of mango?

2 : 5 = 6 : 15

Answer **15** cups

32) 6 bottles of water cost $4.20. How much is 1 bottle?

$4.20 ÷ 6 = $0.70

Answer **$0.70**

Grade 6 Pre-Assessment (continued)

Part 16 Word problems (continued)

33) A school buys 2,016 pencils in boxes of 24. How many boxes?

2,016 ÷ 24 = 84

Answer **84** boxes

34) A $45 shirt is 20% off. How many dollars do you save?

20% of 45 = 9

Answer **$9**

35) You have 3/4 of a pizza. Each slice is 1/8 of the pizza. How many slices is that?

3/4 ÷ 1/8 = 6

Answer **6** slices

Grade 6 Pre-Assessment: Scoring

Count correct answers in each part. Each problem is 1 point.

Part	Skill	Packet	Problems	Score
1	Ratios	91	1–2	___ / 2
2	Rates	92	3–4	___ / 2
3	Percents	93	5–6	___ / 2
4	Dividing fractions	94	7–8	___ / 2
5	Long division	95	9–10	___ / 2
6	Decimal operations	96	11–12	___ / 2
7	GCF and LCM	97	13–14	___ / 2
8	Negative numbers	98	15–16	___ / 2
9	Expressions	99	17–18	___ / 2
10	One-step equations	100	19–20	___ / 2
11	Area of shapes	101	21–22	___ / 2
12	Volume with fractions	102	23–24	___ / 2
13	Surface area	103	25–26	___ / 2
14	Mean and median	104	27–28	___ / 2
15	Data	105	29	___ / 1
16	Word problems	91–105	30–35	___ / 6
	Total			**___ / 35**

Use the pre-assessment to place students: start each student at the first packet where they miss more than 1 problem in its parts. Skip packets they already show mastery on (all or all but one correct).